```
\begin{equation*}
  p_k(x)=\prod_{\substack{i=1\\i\ne k}}^n
              \left(\frac{x-t_i}{t_k-t_i}\right)
\end{equation*}
```

The amsmath package has also a `\sideset` command which can be used to put symbols at any of the four corners of a *large operator*. Thus

`$\sideset{_{ll}^{ul}}{_{lr}^{ur}}\bigcup$` produces $\sideset{_{ll}^{ul}}{_{lr}^{ur}}\bigcup$

`$\sideset{}{'}\sum$` produces $\sideset{}{'}\sum$.

VIII.5. NEW OPERATORS

Mathematical text is usually typeset in italics, and TEX follows this tradition. But certain functions in mathematics such as log, sin, lim and so on are traditionally typeset in roman. This is implemented in TEX by the use of commands like `$\log$`, `$\sin$`, `$\lim$` and so on. The symbols classified as "Log-like symbols" in the table at the end of this chapter shows such functions which are predefined in LATEX.

Having read thus far, it may be no surprise to learn that we can define our own "operator names" which receive this special typographic treatment. This is done by the `\DeclareMathOperator` command. Thus if the operator cl occurs frequently in the document, you can make the declaration

```
\DeclareMathOperator{\cl}{cl}
```

in the preamble and then type `$\cl(A)$` to produce cl(A), for example.

Note that an operator defined like this accommodates subscripts and superscripts in the usual way, that is, at its sides. Thus

```
We denote the closure of $A$ in the subspace $Y$ of $X$ by
$\cl_Y(A)$
```

produces

We denote the closure of A in the subspace Y of X by $\mathrm{cl}_Y(A)$

If we want to define a new operator with subscripts and superscripts placed in the "limits" position below and above, then we should use the starred form of the `\DeclareMathOperator` as shown below

```
\DeclareMathOperator*{\esup}{ess\,sup}

For $f\in L^\infty(R)$, we define
\begin{equation*}
  ||f||_\infty=\esup_{x\in R}|f(x)|
\end{equation*}
```

(Note that the declaration *must* be done in the preamble.) This produces the output

For $f \in L^\infty(R)$, we define

$$\|f\|_\infty = \operatorname*{ess\,sup}_{x\in R} |f(x)|$$

(Why the `\,` command in the definition?)

VIII.6. THE MANY FACES OF MATHEMATICS

We have noted that most mathematics is typeset in italics typeface and some mathematical operators are typeset in an upright fashion. There may be need for additional typefaces as in typesetting vectors in boldface.

LaTeX includes several styles to typeset mathematics as shown in the table below

TYPE STYLE	COMMAND	EXAMPLE	
		INPUT	OUTPUT
italic (default)	`\mathit`	`$x+y=z$`	$x+y=z$
roman	`\mathrm`	`$\mathrm{x+y=z}$`	$\mathrm{x+y=z}$
bold	`\mathbf`	`$\mathbf{x+y=z}$`	$\mathbf{x+y=z}$
sans serif	`\mathsf`	`$\mathsf{x+y=z}$`	$\mathsf{x+y=z}$
typewriter	`\mathtt`	`$\mathtt{x+y=z}$`	$\mathtt{x+y=z}$
calligraphic (upper case only)	`\mathcal`	`$\mathcal{X+Y=Z}$`	$\mathcal{X+Y=Z}$

In addition to these, several other math alphabets are available in various packages (some of which are shown in the list of symbols at the end of this chapter).

Note that the command `\mathbf` produces only *roman* boldface and *not math italic* boldface. Sometimes you may need boldface math italic, for example to typeset vectors. For this, amsmath provides the `\boldsymbol` command. Thus we can get

In this case, we define
$$\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{c}$$

from the input

```
In this case, we define
\begin{equation*}
  \boldsymbol{a}+\boldsymbol{b}=\boldsymbol{c}
\end{equation*}
```

If the document contains several occurrences of such symbols, it is better to make a new definition such as

```
\newcommand{\vect}[1]{\boldsymbol{#1}}
```

and then use `$\vect{a}$` to produce $\boldsymbol{a}$ and `$\vect{b}$` to produce $\boldsymbol{b}$ and so on. the additional advantage of this approach is that if you change your mind later and want vectors to be typeset with arrows above them as $\vec{a}$, then all you need is to change the `\boldsymol` part of the definition of `\vect` to `\overrightarrow` and the change will be effected throughout the document.

Now if we change the input of the above example as

```
In this case, we define
\begin{equation*}
  \boldsymbol{a+b=c}
\end{equation*}
```

then we get the output

In this case, we define

$$\boldsymbol{a + b = c}$$

Note that now the symbols + and = are also in boldface. Thus `\boldsymbol` makes bold every math symbol in its scope (provided the bold version of that symbol is available in the current math font).

There is another reason for tweaking the math fonts. Recently, the International Standards Organization (ISO) has established the recognized typesetting standards in mathematics. Some of the points in it are,

1. Simple variables are represented by italic letters as a, x.
2. Vectors are written in boldface italic as $\boldsymbol{a}$, $\boldsymbol{x}$.
3. Matrices may appear in sans serif as in A, X.
4. The special numbers e, i and the differential operator d are written in *upright roman*.

Point 1 is the default in LaTeX and we have seen how point 2 can be implemented. to fulfill Point 4, it is enough if we define something like

```
\newcommand{\me}{\mathrm{e}}
\newcommand{\mi}{\mathrm{i}}
\newcommand{\diff}{\mathrm{d}}
```

and then use `$\me$` for e and `$\mi$` for i and `$\diff x$` for dx.

Point 3 can be implemented using `\mathsf` but it is a bit difficult (but not impossible) if we need them to be in *italic* also. The solution is to create a new math alphabet, say, `\mathsfsl` by the command

```
\DeclareMathAlphabet{\mathsfsl}{OT1}{cmss}{m}{sl}
```

(in the preamble) and use it to define a command `\matr` to typeset matrices in this font by

```
\newcommand{\matr}[1]{\ensuremath{\mathsfsl{#1}}}
```

so that `$\maqtr A$` produces $\mathsfsl{A}$.

VIII.7. AND THAT IS NOT ALL!

We have only briefly discussed the basic techniques of typesetting mathematics using LaTeX and some of the features of the amsmath package which helps us in this task. For more details on this package see the document `amsldoc.dvi` which should be available with your TeX distribution. If you want to produce *really* beautiful mathematical documents, read the Master—"The TeX Book" by Donald Knuth, especially Chapter 18, "Fine Points of Mathematics Typing".

VIII.8. SYMBOLS

Table VIII.1: Greek Letters

α	`\alpha`	θ	`\theta`	o	`o`	τ	`\tau`
β	`\beta`	ϑ	`\vartheta`	π	`\pi`	υ	`\upsilon`
γ	`\gamma`	ι	`\iota`	ϖ	`\varpi`	ϕ	`\phi`
δ	`\delta`	κ	`\kappa`	ρ	`\rho`	φ	`\varphi`
ϵ	`\epsilon`	λ	`\lambda`	ϱ	`\varrho`	χ	`\chi`
ε	`\varepsilon`	μ	`\mu`	σ	`\sigma`	ψ	`\psi`

ζ	\zeta	ν	\nu	ς	\varsigma	ω	\omega
η	\eta	ξ	\xi				
Γ	\Gamma	Λ	\Lambda	Σ	\Sigma	Ψ	\Psi
Δ	\Delta	Ξ	\Xi	Υ	\Upsilon	Ω	\Omega
Θ	\Theta	Π	\Pi	Φ	\Phi		

Table VIII.2: Binary Operation Symbols

$\pm$	\pm	$\cap$	\cap	$\diamond$	\diamond	$\oplus$	\oplus
$\mp$	\mp	$\cup$	\cup	$\triangle$	\bigtriangleup	$\ominus$	\ominus
$\times$	\times	$\uplus$	\uplus	$\triangledown$	\bigtriangledown	$\otimes$	\otimes
$\div$	\div	$\sqcap$	\sqcap	$\triangleleft$	\triangleleft	$\oslash$	\oslash
$*$	\ast	$\sqcup$	\sqcup	$\triangleright$	\triangleright	$\odot$	\odot
$\star$	\star	$\vee$	\vee	$\lhd$*	\lhd*	$\bigcirc$	\bigcirc
$\circ$	\circ	$\wedge$	\wedge	$\rhd$*	\rhd*	$\dagger$	\dagger
$\bullet$	\bullet	$\setminus$	\setminus	$\unlhd$*	\unlhd*	$\ddagger$	\ddagger
$\cdot$	\cdot	$\wr$	\wr	$\unrhd$*	\unrhd*	$\amalg$	\amalg
$+$	+	$-$	-				

* Not predefined in LaTeX 2ε. Use one of the packages latexsym, amsfonts or amssymb.

Table VIII.3: Relation Symbols

$\leq$	\leq	$\geq$	\geq	$\equiv$	\equiv	$\models$	\models
$\prec$	\prec	$\succ$	\succ	$\sim$	\sim	$\perp$	\perp
$\preceq$	\preceq	$\succeq$	\succeq	$\simeq$	\simeq	$\mid$	\mid
$\ll$	\ll	$\gg$	\gg	$\asymp$	\asymp	$\parallel$	\parallel
$\subset$	\subset	$\supset$	\supset	$\approx$	\approx	$\bowtie$	\bowtie
$\subseteq$	\subseteq	$\supseteq$	\supseteq	$\cong$	\cong	$\bowtie$	\Join*
$\sqsubset$	\sqsubset*	$\sqsupset$	\sqsupset*	$\neq$	\neq	$\frown$	\smile
$\sqsubseteq$	\sqsubseteq	$\sqsupseteq$	\sqsupseteq	$\doteq$	\doteq	$\smile$	\frown
$\in$	\in	$\ni$	\ni	$\propto$	\propto	$=$	=
$\vdash$	\vdash	$\dashv$	\dashv	$<$	<	$>$	>
$:$	:						

* Not predefined in LaTeX 2ε. Use one of the packages latexsym, amsfonts or amssymb.

Table VIII.4: Punctuation Symbols

,	,	;	;	:	\colon	.	\ldotp	·	\cdotp

Table VIII.5: Arrow Symbols

$\leftarrow$	\leftarrow	$\longleftarrow$	\longleftarrow	$\uparrow$	\uparrow
$\Leftarrow$	\Leftarrow	$\Longleftarrow$	\Longleftarrow	$\Uparrow$	\Uparrow
$\rightarrow$	\rightarrow	$\longrightarrow$	\longrightarrow	$\downarrow$	\downarrow
$\Rightarrow$	\Rightarrow	$\Longrightarrow$	\Longrightarrow	$\Downarrow$	\Downarrow
$\leftrightarrow$	\leftrightarrow	$\longleftrightarrow$	\longleftrightarrow	$\updownarrow$	\updownarrow
$\Leftrightarrow$	\Leftrightarrow	$\Longleftrightarrow$	\Longleftrightarrow	$\Updownarrow$	\Updownarrow
$\mapsto$	\mapsto	$\longmapsto$	\longmapsto	$\nearrow$	\nearrow

↩	\hookleftarrow	↪	\hookrightarrow	↘	\searrow
↼	\leftharpoonup	⇀	\rightharpoonup	↙	\swarrow
↽	\leftharpoondown	⇁	\rightharpoondown	↖	\nwarrow
⇌	\rightleftharpoons	⇝	\leadsto*		

* Not predefined in LaTeX 2ε. Use one of the packages latexsym, amsfonts or amssymb.

Table VIII.6: Miscellaneous Symbols

…	\ldots	⋯	\cdots	⋮	\vdots	⋱	\ddots
ℵ	\aleph	′	\prime	∀	\forall	∞	\infty
ℏ	\hbar	∅	\emptyset	∃	\exists	□	\Box*
ı	\imath	∇	\nabla	¬	\neg	◇	\Diamond*
ȷ	\jmath	√	\surd	♭	\flat	△	\triangle
ℓ	\ell	⊤	\top	♮	\natural	♣	\clubsuit
℘	\wp	⊥	\bot	♯	\sharp	♢	\diamondsuit
ℜ	\Re	∥	\\|	\	\backslash	♡	\heartsuit
ℑ	\Im	∠	\angle	∂	\partial	♠	\spadesuit
℧	\mho*	.	.	|	|		

* Not predefined in LaTeX 2ε. Use one of the packages latexsym, amsfonts or amssymb.

Table VIII.7: Variable-sized Symbols

∑	\sum	⋂	\bigcap	⨀	\bigodot
∏	\prod	⋃	\bigcup	⨂	\bigotimes
∐	\coprod	⨆	\bigsqcup	⨁	\bigoplus
∫	\int	⋁	\bigvee	⨄	\biguplus
∮	\oint	⋀	\bigwedge		

Table VIII.8: Log-like Symbols

\arccos	\cos	\csc	\exp	\ker	\limsup	\min	\sinh
\arcsin	\cosh	\deg	\gcd	\lg	\ln	\Pr	\sup
\arctan	\cot	\det	\hom	\lim	\log	\sec	\tan
\arg	\coth	\dim	\inf	\liminf	\max	\sin	\tanh

Table VIII.9: Delimiters

(	(	)	)	↑	\uparrow	⇑	\Uparrow
[	[	]	]	↓	\downarrow	⇓	\Downarrow
{	\{	}	\}	↕	\updownarrow	⇕	\Updownarrow
⌊	\lfloor	⌋	\rfloor	⌈	\lceil	⌉	\rceil
⟨	\langle	⟩	\rangle	/	/	\	\backslash
|	|	‖	\\|			

Table VIII.10: Large Delimiters

⎰	\rmoustache	⎱	\lmoustache	⟯	\rgroup	⟮	\lgroup
⏐	\arrowvert	⏐	\Arrowvert	⏐	\bracevert		

Table VIII.11: Math Mode Accents

$\hat{a}$	\hat{a}	$\acute{a}$	\acute{a}	$\bar{a}$	\bar{a}	$\dot{a}$	\dot{a}
$\breve{a}$	\breve{a}	$\check{a}$	\check{a}	$\grave{a}$	\grave{a}	$\vec{a}$	\vec{a}
$\ddot{a}$	\ddot{a}	$\tilde{a}$	\tilde{a}				

Table VIII.12: Some other Constructions

$\widetilde{abc}$	\widetilde{abc}	$\widehat{abc}$	\widehat{abc}
$\overleftarrow{abc}$	\overleftarrow{abc}	$\overrightarrow{abc}$	\overrightarrow{abc}
$\overline{abc}$	\overline{abc}	$\underline{abc}$	\underline{abc}
$\overbrace{abc}$	\overbrace{abc}	$\underbrace{abc}$	\underbrace{abc}
$\sqrt{abc}$	\sqrt{abc}	$\sqrt[n]{abc}$	\sqrt[n]{abc}
f'	f'	$\frac{abc}{xyz}$	\frac{abc}{xyz}

Table VIII.13: AMS Delimiters

⌜	\ulcorner	⌝	\urcorner	⌞	\llcorner	⌟	\lrcorner

Table VIII.14: AMS Arrows

⇢	\dashrightarrow	⇠	\dashleftarrow
⇇	\leftleftarrows	⇆	\leftrightarrows
⇚	\Lleftarrow	↞	\twoheadleftarrow
↢	\leftarrowtail	↫	\looparrowleft
⇋	\leftrightharpoons	↶	\curvearrowleft
↺	\circlearrowleft	↰	\Lsh
⇈	\upuparrows	↿	\upharpoonleft
⇃	\downharpoonleft	⊸	\multimap
↭	\leftrightsquigarrow	⇉	\rightrightarrows
⇄	\rightleftarrows	⇉	\rightrightarrows
⇄	\rightleftarrows	↠	\twoheadrightarrow
↣	\rightarrowtail	↬	\looparrowright
⇌	\rightleftharpoons	↷	\curvearrowright
↻	\circlearrowright	↱	\Rsh
⇊	\downdownarrows	↾	\upharpoonright
⇂	\downharpoonright	⇝	\rightsquigarrow

Table VIII.15: AMS Negated Arrows

↚	\nleftarrow	↛	\nrightarrow	⇍	\nLeftarrow
⇏	\nRightarrow	↮	\nleftrightarrow	⇎	\nLeftrightarrow

Table VIII.16: AMS Greek

F	\digamma	$\varkappa$	\varkappa

Table VIII.17: AMS Hebrew

ב	\beth	ד	\daleth
ג	\gimel		

Table VIII.18: AMS Miscellaneous

$\hbar$	\hbar	$\hbar$	\hslash
□	\square	◊	\lozenge
∡	\measuredangle	∄	\nexists
⅁	\Game	𝕜	\Bbbk
▲	\blacktriangle	▼	\blacktriangledown
★	\bigstar	∢	\sphericalangle
⁄	\diagup	╲	\diagdown
△	\vartriangle	▽	\triangledown
Ⓢ	\circledS	∠	\angle
℧	\mho	ⅎ	\Finv
‵	\backprime	∅	\varnothing
■	\blacksquare	⧫	\blacklozenge
∁	\complement	ð	\eth

Table VIII.19: AMS Binary Operators

∔	\dotplus	∖	\smallsetminus	⋒	\Cap
⩒	\barwedge	⩔	\veebar	⩓	\doublebarwedge
⊠	\boxtimes	⊡	\boxdot	⊞	\boxplus
⋉	\ltimes	⋊	\rtimes	⋋	\leftthreetimes
⋏	\curlywedge	⋎	\curlyvee	⊖	\circleddash
◎	\circledcirc	⋅	\centerdot	⊺	\intercal
⋓	\Cup	⊟	\boxminus	⋇	\divideontimes
⋌	\rightthreetimes	⊛	\circledast		

Table VIII.20: AMS Binary Relations

≦	\leqq	⩽	\leqslant	⪕	\eqslantless
≲	\lessapprox	≈	\approxeq	⋖	\lessdot
≶	\lessgtr	⋚	\lesseqgtr	⪋	\lesseqqgtr
≒	\risingdotseq	≓	\fallingdotseq	∽	\backsim
⫅	\subseteqq	⋐	\Subset	⊏	\sqsubset
≼	\curlyeqprec	≾	\precsim	⪷	\precapprox
⊴	\trianglelefteq	⊨	\vDash	⊪	\Vvdash
⌢	\smallfrown	≏	\bumpeq	≎	\Bumpeq
⩾	\geqslant	⪖	\eqslantgtr	≳	\gtrsim
⪀	\gtrdot	⋙	\ggg	≷	\gtrless
⪌	\gtreqqless	≖	\eqcirc	≗	\circeq
∼	\thicksim	≈	\thickapprox	⫆	\supseteqq

⊐	\sqsupset	≽	\succcurlyeq	⋟	\curlyeqsucc
≿	\succapprox	⊳	\vartriangleright	⊵	\trianglerighteq
∣	\shortmid	∥	\shortparallel	≬	\between
∝	\varpropto	◀	\blacktriangleleft	∴	\therefore
▶	\blacktriangleright	∵	\because	≲	\lesssim
⋘	\lll	≒	\doteqdot	≏	\backsimeq
≼	\preccurlyeq	◁	\vartriangleleft	⌣	\smallsmile
≧	\geqq	≳	\gtrapprox	≷	\gtreqless
≜	\triangleq	⋑	\Supset	≳	\succsim
⊩	\Vdash	⋔	\pitchfork	∋	\backepsilon

Table VIII.21: AMS Negated Binary Relations

≮	\nless	≰	\nleq	⩽̸	\nleqslant
≨	\lneq	≨	\lneqq	≨	\lvertneqq
⋦	\lnapprox	⊀	\nprec	⋠	\npreceq
⋨	\precnapprox	≁	\nsim	∤	\nshortmid
⊬	\nvdash	⊭	\nvDash	⋪	\ntriangleleft
⊈	\nsubseteq	⊊	\subsetneq	⊊	\varsubsetneq
⫋	\varsubsetneqq	≯	\ngtr	≱	\ngeq
≱	\ngeqq	≩	\gneq	≩	\gneqq
≵	\gnsim	⋧	\gnapprox	⊁	\nsucc
⋡	\nsucceq	⋩	\succnsim	⋧	\succnapprox
∦	\nshortparallel	∦	\nparallel	⊯	\nvDash
⋫	\ntriangleright	⋭	\ntrianglerighteq	⊉	\nsupseteq
⊋	\supsetneq	⊋	\varsupsetneq	⫌	\supsetneqq
≦̸	\nleqq	≴	\lnsim	⋨	\precnsim
∤	\nmid	⋪	\ntrianglelefteq	⫋	\subsetneqq
⩾̸	\ngeqslant	≩	\gvertneqq	⊁	\nsucceq
≇	\ncong	⊯	\nVDash	⊉	\nsupseteqq
⊋	\varsupsetneqq				

Table VIII.22: Math Alphabets

		Required package
ABCdef	\mathrm{ABCdef}	
ABCdef	\mathit{ABCdef}	
ABCdef	\mathnormal{ABCdef}	
$\mathcal{ABC}$	\mathcal{ABC}	
$\mathcal{ABC}$	\mathcal{ABC}	euscript with option: mathcal
	\mathscr{ABC}	euscript with option: mathcr
$\mathfrak{ABCdef}$	\mathfrak{ABCdef}	eufrak
$\mathbb{ABC}$	\mathbb{ABC}	amsfonts or amssymb
$\mathscr{ABC}$	\mathscr{ABC}	mathrsfs